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Subhalingam D

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Education

Indian Institute of Technology, Delhi **Jul 2018 – May 2022**
Bachelor of Technology in Mathematics and Computing | *CGPA: 8.196/10*

Experience

Data Scientist | **KnowDis AI, Delhi** **May 2022 – Present**

Product Category Prediction for Automated Product Mapping (for a Large B2B Marketplace)

- Designed a retriever-reranker pipeline to classify products into one of **100K+ categories** using metadata like title and specifications
- Curated training data from a **120M+ product catalog** and augmented it with task-specific heuristics to improve data quality
- Fine-tuned a decoder-based retriever; aggregated candidate rankings using **Reciprocal Rank Fusion** across multiple retrievers
- Developed few-shot prompts for Qwen LLM reranker for single-step decoding; optimized latency using prefix-caching and **vLLM**
- Achieved **13% accuracy gain**, reduced grossly wrong predictions by 30%, and maintained 92%+ high-confidence coverage
- Integrated the end-to-end system; led the dockerization and coordinated client delivery & testing; solution **deployed to production**

Attribute-Value Extraction from Product Titles & User Queries (for a Large B2B Marketplace) *(Research accepted at KDD 2025)*

- Generated weakly supervised training data with **incomplete labeling** from product specifications covering 25K+ attributes
- Architected a novel two-stage system that employs a **marker-augmented** generative model to identify potential attributes, followed by a cross-encoder-based token classification model that determines the associated values for each attribute
- **Regenerated training data** to expand attribute-value annotations and **trained a faster NER classifier** on the enriched dataset
- Improved **recall by 20%** while maintaining 90% precision; used for dynamic feature highlighting to enhance search experience

Additional Contributions Across ML, NLP & Generative AI

- **Query Understanding Layer for Product Search:** Worked on low-latency spell correction, romanized Indian-language translation and query intent classification models to improve downstream search pipeline performance, serving 2M+ unique queries weekly
- **Product Listing Auditor:** Orchestrated an agentic system to streamline seller journey by automating product audits and approvals
- **Visual Product Search Engine:** Introduced image-based search, serving 1M+ requests to date (at p50 < 2.5s); developing features for conversational and multimodal composed retrieval to support attribute edits, complementary products, and search refinement
- **Style-Controllable Translation (En-Hi):** Fine-tuned an encoder-decoder model on in-house curated corpora to control formality
- **Temporal RAG System:** Building recency-aware rescoring for a Q&A chatbot to retrieve up-to-date info for time-sensitive queries
- **AI for Drug Discovery:** Developed *in-silico* methods for *de novo* molecule generation, virtual screening and target deconvolution

Software Engineering Intern | **Samsung R&D Institute, Delhi** **Jun 2021 – Jul 2021**
Acoustic Sound Source Localization, Tracking and Separation | *Received Pre-Placement Offer (PPO)*

Skills

Languages:	**Python**, Java, C++, C, SQL, Bash
AI/ML & NLP:	**PyTorch**, **Transformers**, Elasticsearch, FAISS, LLMs, RAG, TRL, NLTK, spaCy, Scikit-learn
Inference & Backend:	**FastAPI**, **AWS**, **NVIDIA Triton**, **vLLM**, ONNX, LangChain, LangGraph, Streamlit, Celery, Docker, Git

Publications

1. **A Framework for Leveraging Partially-Labeled Data for Product Attribute-Value Identification** | **D. Subhalingam**, Keshav Kolluru, Mausam & Saurabh Singal | **KDD 2025, Industry Track** *(Acceptance Rate: 22%)*
2. **TAGMol: Target-Aware Gradient-guided Molecule Generation** | Vineeth Dorna, **D. Subhalingam**, Keshav Kolluru, Shreshth Tuli, Mrityunjay Singh, Saurabh Singal, N. M. Anoop Krishnan & Sayan Ranu | **ICML 2024 Workshop on ML for Life and Material Science**

Projects

- **Tracking Entities in Technical Procedural Text** | Prof. Srikanta Bedathur & Prof. Maya Ramanath **Feb 2021 – Apr 2022**
  Curated a WikiHow-based dataset and built BERT & WordNet baselines to predict entity state changes across procedural steps
- **Identification of Hate Spreaders on Social Media** | Prof. Niladri Chatterjee **Jan 2022 – Apr 2022**
  Proposed GloVe + sentiment-weighted scheme to profile hate spreaders on Twitter; 76% accuracy on PAN@CLEF 2021 (+1% over best)
- **Rule-based Written-to-Spoken Text Converter** | Prof. Mausam **Aug 2021 – Sep 2021**
  Built a rule-based system to rewrite text in spoken form, handling dates, abbreviations and units; achieved 97.94% F1 (top in class)
- **Sparse Retrieval with Extended Vector Space Model** | Prof. Srikanta Bedathur **Oct 2020 – Nov 2020**
  Built a TF-IDF retrieval system with prefix search & gap encoding (halved index size); explored pseudo-relevance feedback reranking

Activities

- Invited **Reviewer**, KDD 2026 Datasets and Benchmarks Track (August); **Emergency Reviewer**, ACL Rolling Review - February 2025
- **General Secretary,** Mathematics Society, IIT Delhi Aug 2021 – Jun 2022
- **Teaching Assistant,** Information Retrieval and Web Search *(Graduate course, Prof. Srikanta Bedathur)* Aug 2021 – Dec 2021